Principles of bionics and biorobotics engineering

Bioelectronics and neuroengineering in neurology and neurorehabilitation – Part 3

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3. Technologies for functional assessment

3.1 Clinical assessment

- Overview
- Assessment in neurology and neurorehabilitation

3.2 Technologies for measuring human motion

- Optical motion capture systems
- Wearable sensors
- Electromyography

3.3 Emerging techniques for monitoring and assessment

- Telemonitoring applications
- Prediction of disease course and rehabilitation outcomes

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Clinical assessment

Clinical assessments are essential to **monitor how neurological or neurodegenerative disorders evolve** over time. By quantifying motor impairments, clinicians can identify patterns of decline or stability, which helps in determining the stage of the disease. **Tracking progression** aids in making informed decisions about the timing and intensity of interventions, as well as preparing for anticipated challenges.

Clinical assessments are used to objectively **measure the effectiveness of various treatment modalities**, including physical therapy, medications, surgical interventions, and neurostimulation techniques. They help **quantify improvements**, such as increased motor control, reduced spasticity, or enhanced coordination, providing evidence for the success or need for adjustment of the chosen approach. Comparisons across different interventions can guide the selection of optimal therapies for specific conditions.

Each patient presents unique functional deficits; assessments enable clinicians to identify these individual challenges. **Tailored interventions can be designed to target specific impairments**, such as weakness, tremor, or impaired gait. Personalization ensures that therapy is relevant and effective, maximizing the potential for recovery or symptom management while minimizing unnecessary efforts on aspects that may not benefit the patient.

Clinical assessment

Clinical assessment is critical for monitoring disease progression and evaluating the effectiveness of neurorehabilitation interventions. Focuses primarily on motor functions, including movement coordination, strength, and voluntary control. Applicable in various conditions:

Parkinson's Disease (PD): Tremor, bradykinesia, and rigidity.

Spinal Cord Injury (SCI): Muscle activation, voluntary control, and spasticity.

Stroke: Motor recovery, balance, and gait.

Amyotrophic Lateral Sclerosis (ALS): Muscle weakness and spasticity.

Multiple Sclerosis (MS): Coordination, fatigue, and motor deficits.

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Spinal Cord Injury (SCI): Muscle activation, voluntary control, and spasticity. → American Spinal Injury Association (ASIA) Impairment Scale.

Stroke: Motor recovery, balance, and gait. → Fugl-Meyer Assessment, Barthel Index.

Amyotrophic Lateral Sclerosis (ALS): Muscle weakness and spasticity. → ALS Functional Rating Scale (ALSFRS).

Multiple Sclerosis (MS): Coordination, fatigue, and motor deficits. → Expanded Disability Status Scale (EDSS).

Clinical assessment

Limitations of clinical assessment scales:

- **Subjectivity:** heavily reliant on clinician expertise and interpretation.
- **Snapshot nature:** provide static evaluations, not continuous monitoring.
- **Limited specificity:** May not capture subtle changes in motor function.
- **Time-consuming:** Require significant time investment for both clinicians and patients.

Characteristics of domains of outcome in modified International Classification of Functioning, Disability, and Health model.

Domain	Level of Operation	Example	Can Use Adaptive Aids?	Can Use Another Person's Assistance?
Impairment	Organ system	Strength of finger flexors	No	No
Functional Limitation	Individual as whole	Grasp cylindrical object	No	No
Activity				
Capacity	Individual in standardized environment	Drink from 6 oz glass without handle	Preferably not	Preferably not
Performance	Individual in usual environment	Get drink (any method, i.e., cup, glass, straw)	Yes	Yes
Participation	Individual fulfilling life roles in usual environment	Dine out with friends	Yes	Yes

CAN THEY?

DO THEY?

Marino, R. J. (2007). Domains of outcomes in spinal cord injury for clinical trials to improve neurological function. *Journal of Rehabilitation Research & Development*, *44*(1)

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CAN THEY?

DO THEY?

Need for objective tools: Incorporate motion capture systems, wearable devices, and sEMG.

Continuous monitoring: Shift from clinic-based snapshots to real-time, daily-life data collection.

Integration of technology: Leverage AI and machine learning for personalized insights and predictive analytics.

Marino, R. J. (2007). Domains of outcomes in spinal cord injury for clinical trials to improve neurological function. *Journal of Rehabilitation Research & Development*, *44*(1)

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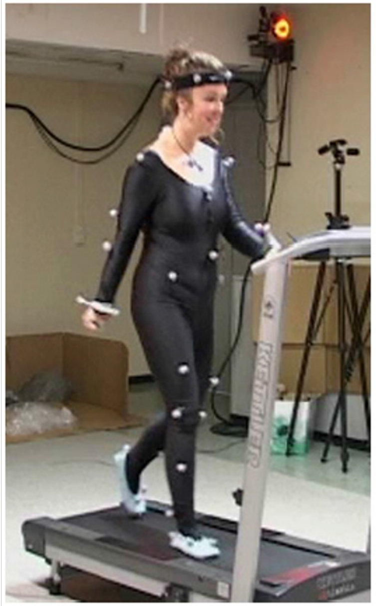
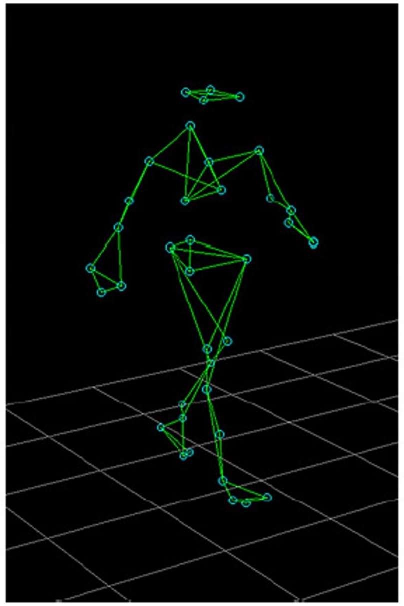
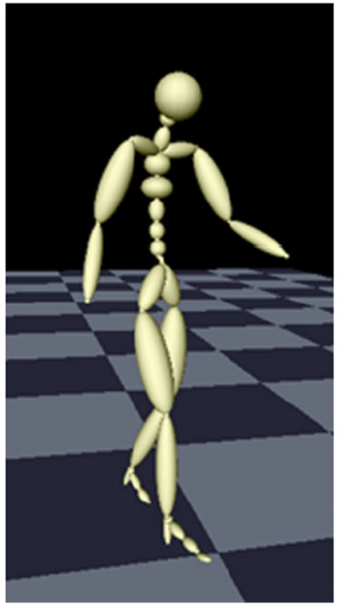
Motion capture systems

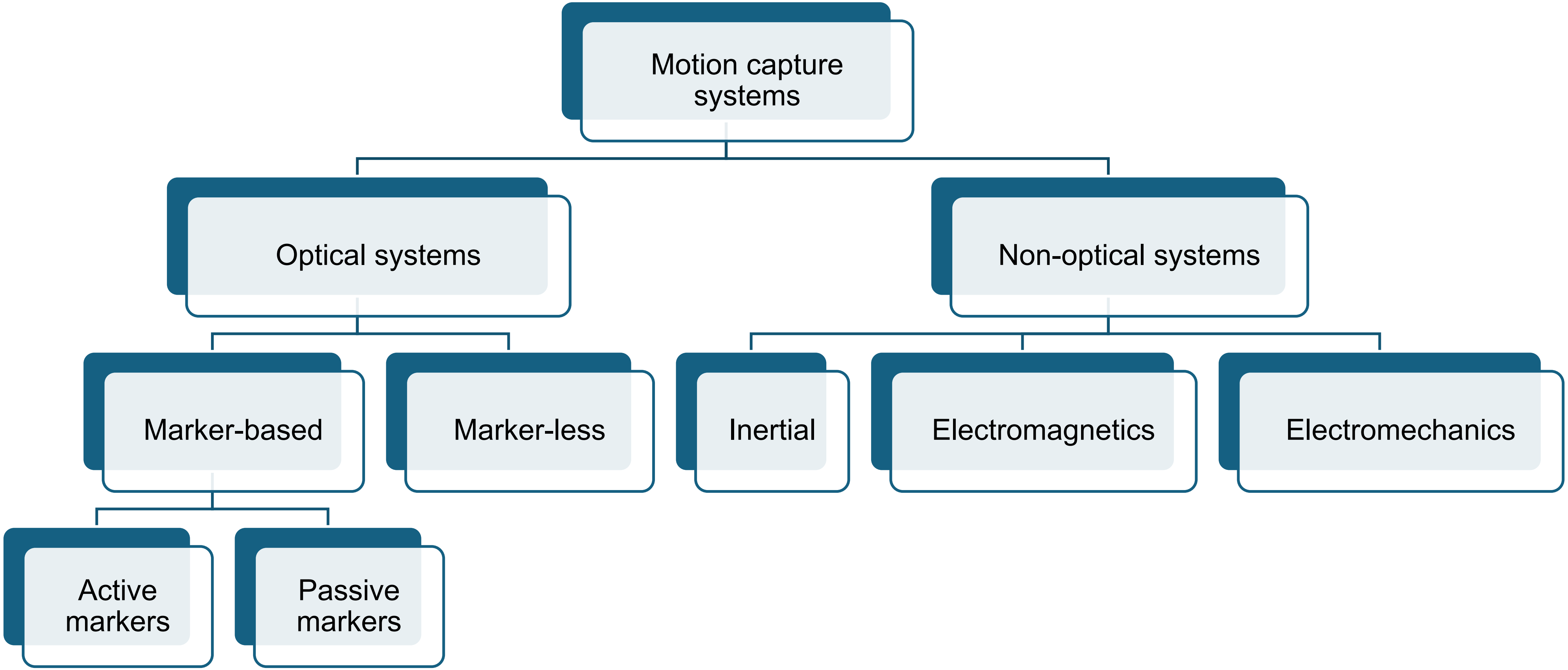
Motion capture (sometimes referred to as mo-cap or mocap, for short) is the process of recording the movements of objects or people.






Motion capture systems

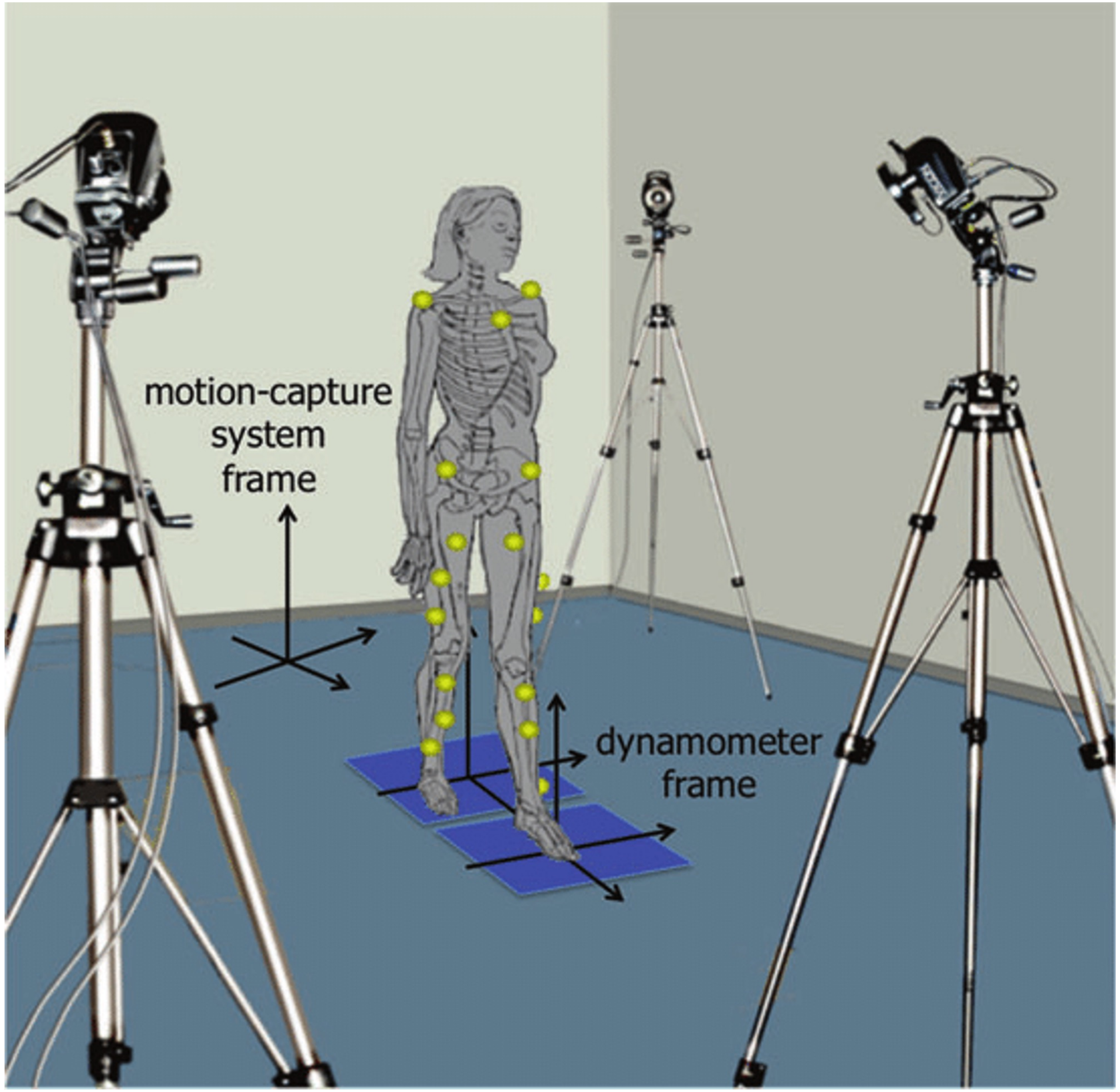


Marker-based mocap

- The most common method for collecting kinematic data uses multiple cameras to record the motion of **markers** affixed to a moving subject.

- 3-D global coordinates of each marker can be estimated from 2D camera coordinates.

- These coordinates are then processed to obtain the kinematic variables that describe segmental or joint movements.



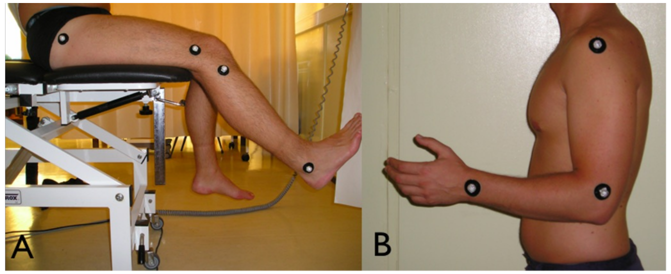
Active vs passive markers

- **Active optical markers**: Markers (usually LEDs) emit infra-red light using different firing sequences that are activated by an electrical current (including wireless).



- **Passive optical markers**: Retro-reflective spheres are illuminated and detected by the infra-red cameras. The spheres are attached to a rigid body with a unique geometry for each tracked device. Markers are illuminated using IR lights mounted on the cameras.

Usually, from 3 mm to 30 mm




Active vs passive markers

Active marker systems

Features
- High-resolution, high-speed cameras
- 120-240 Hz, 1000x1000 pixels
- Infrared or visible light strobe
- Active switching markers

Pros
- High quality
- Flexible marker placement
- No correspondence required

Cons
- Extensive post-processing
- Connection among LEDs
- Structured environments
- Occlusion
- Soft-tissue artifacts

Passive marker systems

Features
- High-resolution, high-speed cameras
- 120-240 Hz, 1000x1000 pixels
- Infrared or visible light strobe
- Reflective markers

Pros
- High quality
- Flexible marker placement
- Not seriously constrained by markers

Cons
- Extensive preparation post-processing
- Correspondence
- Structured environments (indoor only)
- Occlusion
- Soft-tissue artifacts

Motion capture systems

Ideal requirements of a motion capture system:

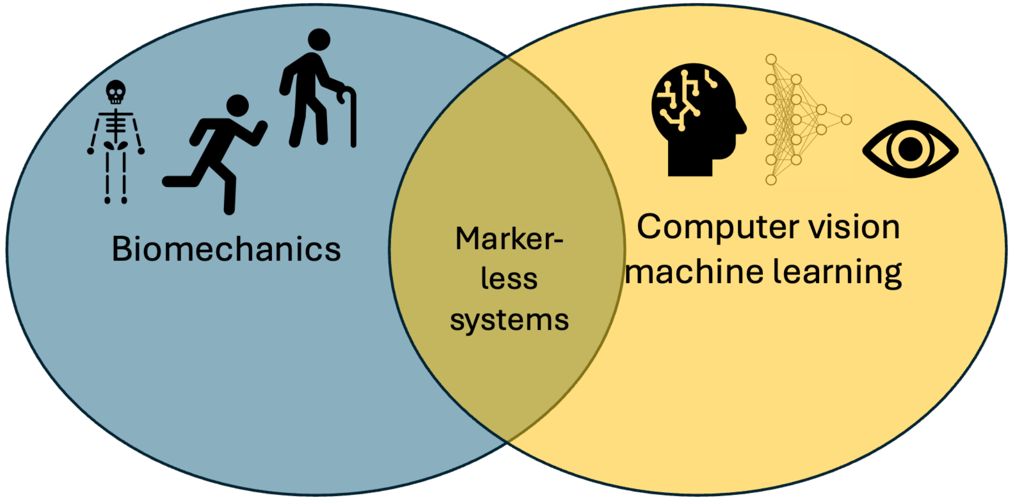
- Accuracy
- Non-invasiveness
- Non-intrusiveness
- Posing minimal risk or discomfort to the subject
- Ability to measure motion in natural environments

Marker-less mocap

Removing the requirement for markers would significantly decrease patient preparation time and facilitate straightforward, time-saving, and potentially more insightful evaluations of human movement in both research and clinical settings.

Nonetheless, obtaining information as precise as marker-based systems remains difficult

- Do not require special sensors or a special suit
- Tracking with common cameras, webcams, or depth cameras
- Few companies provide commercial systems
- Unclear what accuracies these systems can achieve
- Very rapid development (based on technologies of computer vision and pattern recognition)



Marker-less mocap

Hardware and devices

Cameras

Depth sensors

→

Computer vision models

Body pose

Hand pose

Face alignment

→

Analysis and interpretation

Clinical assessment

Gesture recognition

Action/activity detection

Marker-less mocap

Based on the type of **hardware**, systems can be divided into:

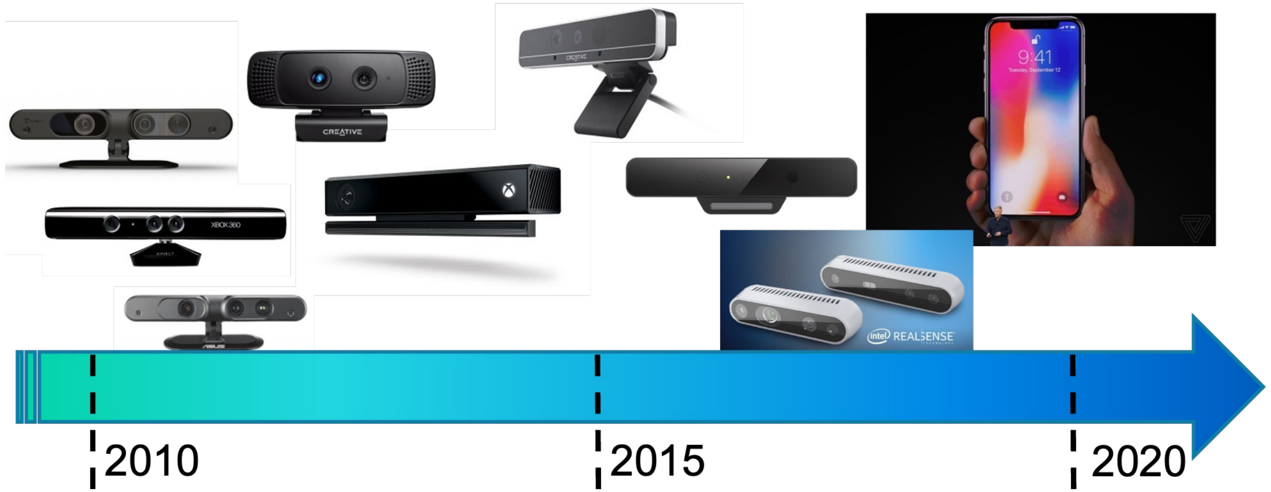
- **Active systems** – they emit light information in the visible or infrared light spectrum in the form of laser light, light patterns or modulated light pulses
- **Passive systems** - they rely purely on capturing color images of the observed scene

Based on the **number of cameras,** systems can be divided into:

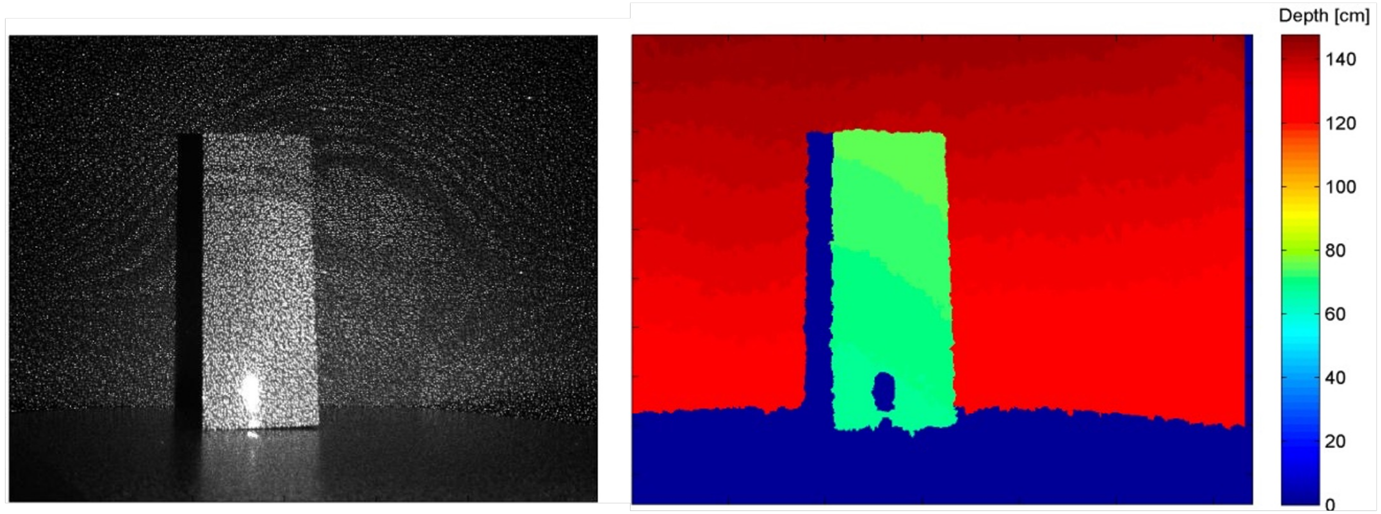
- **Single camera systems** – they can rely on recent computer vision algorithms able to infer the 3D coordinates from 2D planar images
- **Multiple camera systems** – they exploit the principles of 3D triangulation for inferring the 3D coordinates of the points

Marker-less mocap

Hardware and devices



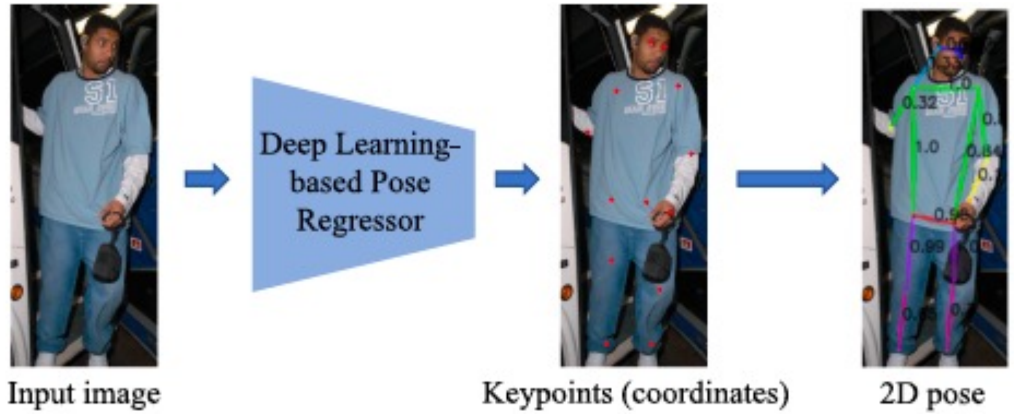
Device	Range	FPS (max)	Price range
Intel RealSense D405	0.07 – 0.5 m	90 fps	200-300 $
Intel RealSense D455	0.6 – 6 m	90 fps	400-500 $
Orbbec Astra Pro Plus	0.6 – 8 m	30 fps	100-200 $
Stereolabs ZED-X	0.5 – 20 m	120 fps	500-1100 $



Marker-less mocap

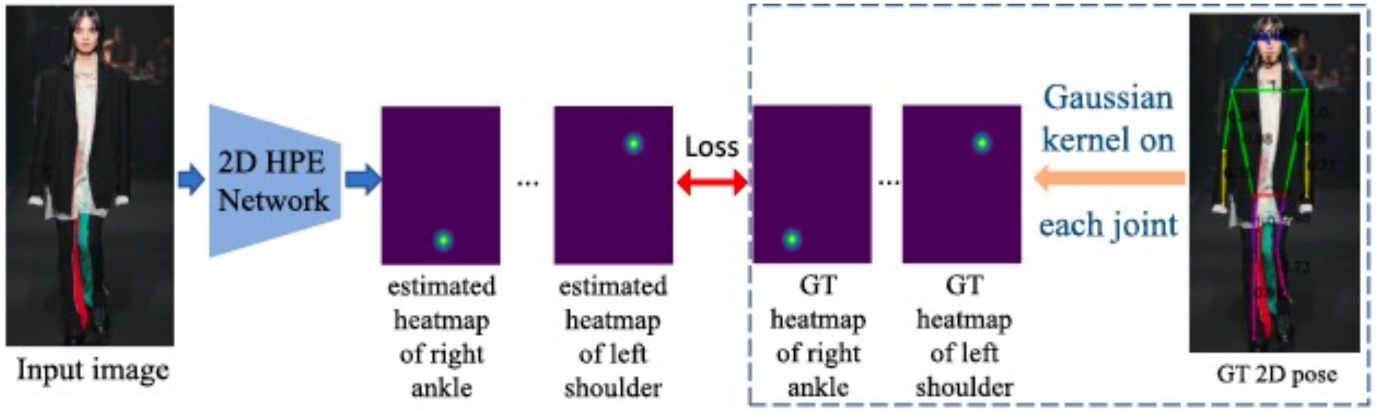
Computer vision models

Regression-based methods
Regression methods directly learn a mapping (via a deep neural network) from the original image to the kinematic body model and produce joint coordinates.



(a) Regression Methods

Heatmap-based methods
Given the ground-truth 2D pose, the ground-truth heatmaps of each joint are generated by applying a Gaussian kernel to each joint's location. Then, heatmap-based methods utilize a model to predict the heatmap of each joint.



(b) Heatmap-based Methods

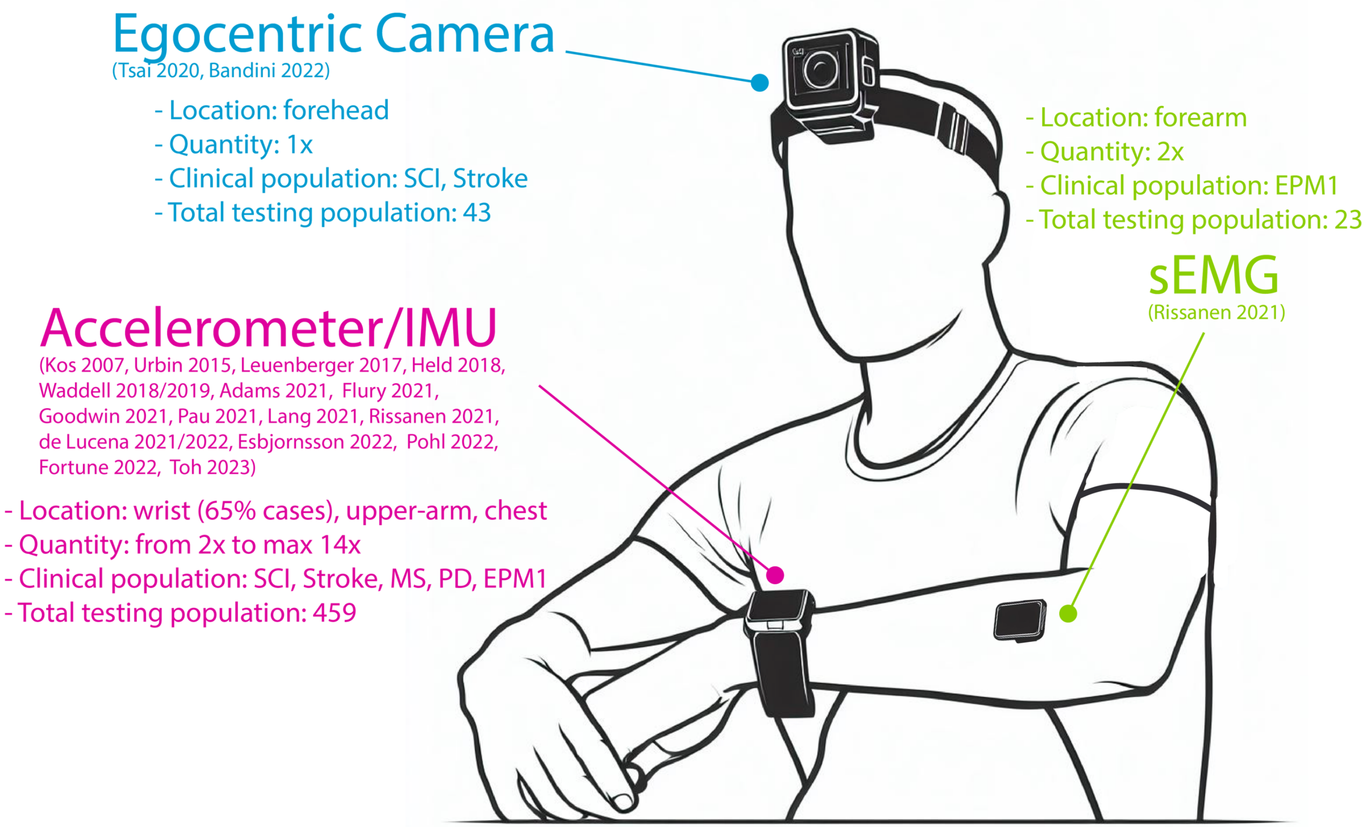
Marker-less mocap

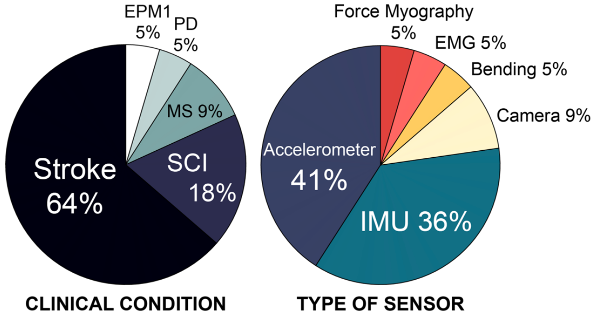
https://github.com/CMU-Perceptual-Computing-Lab/openpose

Cao, Z., Simon, T., Wei, S. E., & Sheikh, Y. (2017). Realtime multi-person 2d pose estimation using part affinity fields. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 7291-7299).

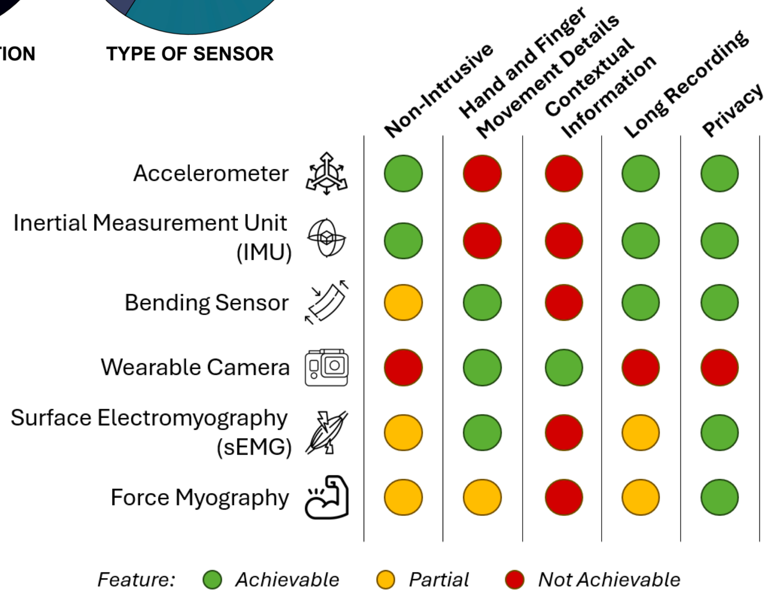
Wearable sensors

In these settings, the use of wearable sensors to study human motion is preferable





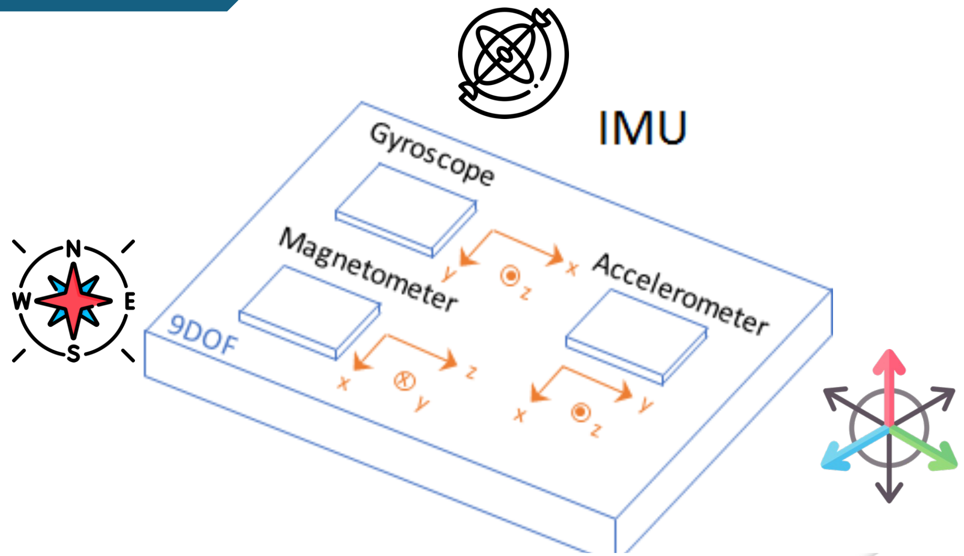
EPM1: Progressive myoclonic epilepsy type 1; **PD**: Parkinson's disease; **SCI**: Spinal cord injury; **MS**: Multiple sclerosis; **IMU**: Inertial measurement unit; **EMG**: Electromyography



Proietti, T., & Bandini, A. (2024). Wearable technologies for monitoring upper extremity functions during daily life in neurologically impaired individuals. *IEEE transactions on neural systems and rehabilitation engineering*.

Inertial sensors

An Inertial Measurement Unit (IMU) includes a combination of individual sensors, including a gyroscope, an accelerometer, and a magnetometer.



IMUs are commonly used as orientation sensors in various consumer products.

- Nearly all smartphones and tablets integrate IMUs for orientation sensing.
- Fitness trackers and other wearables may incorporate IMUs to track motion, such as during running.

Technical Specifications

Device Characteristics

Dimensions: 48.5x36.5x13.5 mm
Weight: 22 grams (with battery)
Material: 6061 clear anodized Aluminum, ABS plastic, and Velcro straps.
Internal Storage: 8 GB (~720 hours of recording)
Battery Life: Wireless Streaming: 8 hours
Synchronous Logging: 12 hours
Asynchronous Logging: 16-36 hours (depending on output rate)

Sensor Characteristics

Property	*Accelerometer*	*Gyroscope*	*Magnetometer*
Axes	3	3	3
Range	+/-2g or +/-6g	+/- 2000 °/s	+/- 6 Gauss
Noise Density	128 µg/√Hz	0.07 °/s/√Hz	4 mGauss/√Hz
Sample Rate	1280 Hz	1280 Hz	1280 Hz
Output Rate	20 - 128 Hz	20 - 128 Hz	20 - 128 Hz
Bandwidth	50 Hz	50 Hz	50 Hz
Resolution	14 bits	14 bits	14 bits

https://shop.movella.com/entertainment/products

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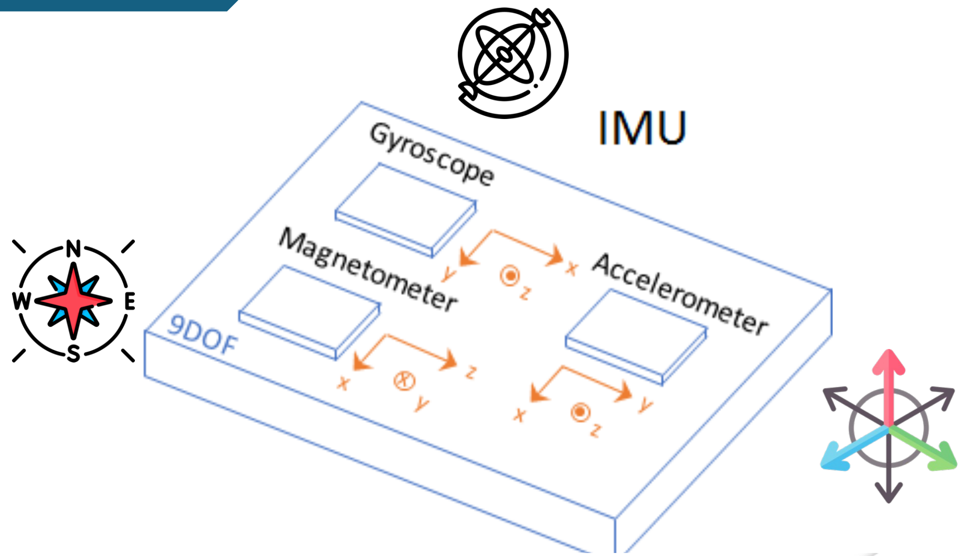

Pros
- Compact and lightweight
- Low cost
- Real-time data
- Versatility
- Minimal setup

Cons
- Measurement drift
- Orientation ambiguity
- Limited accuracy
- Sensor synchronization
- Sensor placement

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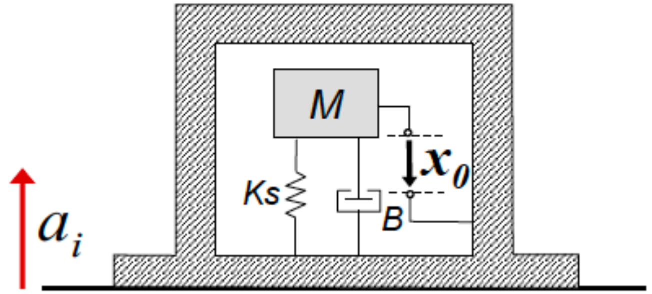
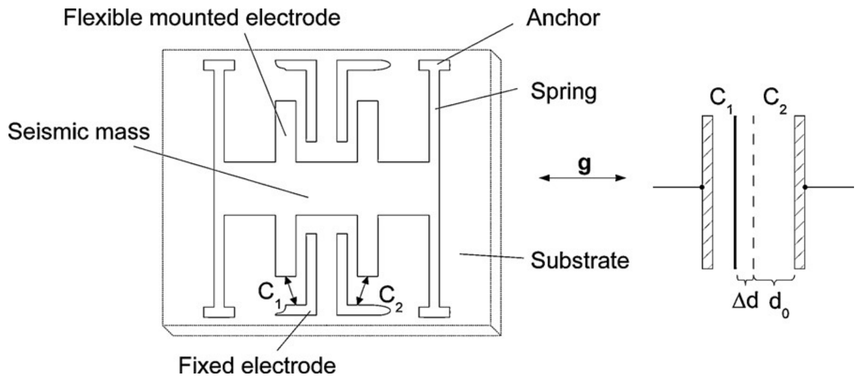
Inertial sensors

Accelerometer – used to sense the linear acceleration



Accelerometers typically use the principles of inertia to measure acceleration. Inside the accelerometer, there is a mass suspended on springs. When the accelerometer experiences acceleration, this mass tends to resist changes in its motion due to inertia.



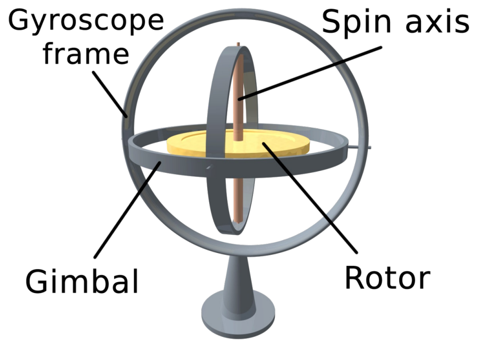
Capacitive sensing: The mass forms one plate of a capacitor, and the casing of the accelerometer forms the other plate. As the mass moves in response to acceleration, the distance between the plates changes, altering the capacitance of the capacitor. This change in capacitance can be measured and used to calculate the acceleration.

Output signals: electrical voltage or digital signal that corresponds to the acceleration experienced along one or more axes.

By measuring acceleration along multiple axes (e.g., x, y, and z), the accelerometer can provide information about the direction and magnitude of acceleration in three-dimensional space.

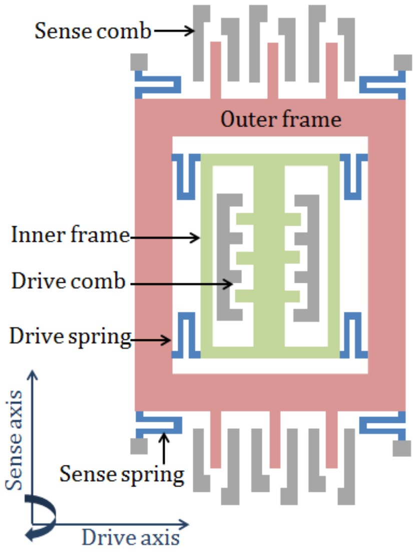
Büsching, F., Kulau, U., Gietzelt, M., & Wolf, L. (2012). Comparison and validation of capacitive accelerometers for health care applications. *Computer methods and programs in biomedicine*, *106*(2), 79-88.

Inertial sensors



Gyroscope – used to sense the angular velocity

Traditional mechanical gyroscopes consist of a spinning rotor mounted on a set of gimbals, allowing it to rotate freely in all directions. When the gyroscope rotates around the axis (y) perpendicular to the spinning mass (x), an angular momentum is developed around the z-axis and can be sensed by torque or force sensor.



Microelectromechanical systems (MEMS) gyroscopes are miniature versions of mechanical gyroscopes fabricated using semiconductor manufacturing techniques. They typically consist of a vibrating mass (proof mass) suspended on a set of flexible beams. When the gyroscope rotates, the Coriolis effect causes the proof mass to deflect, generating a measurable signal.

Output signals: electrical voltage corresponding to the angular velocity around one or more axes.

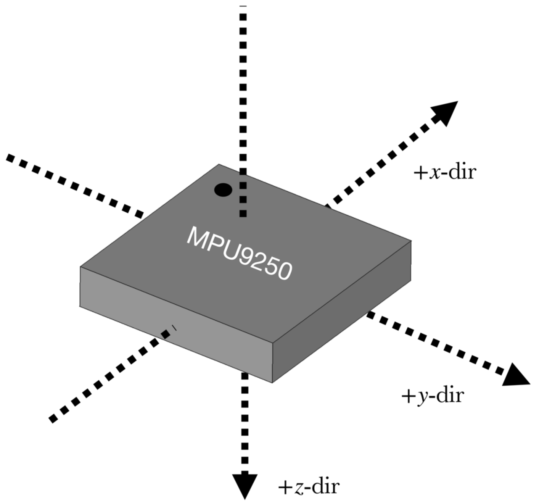
By measuring rotation along multiple axes (e.g., pitch, yaw, and roll), the gyroscope provides information about the orientation and movement of an object in three-dimensional space.

Patel, C., & McCluskey, P. (2012, May). Modeling and simulation of the MEMS vibratory gyroscope. In *13th InterSociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems* (pp. 928-933). IEEE.

Inertial sensors

Magnetometer – used to detect the magnetic field

Magnetometer (AK8963)



Used to determine the sensor's orientation relative to the Earth's magnetic North. The magnetometer can detect the direction of the Earth's magnetic field. As the IMU moves or rotates, the magnetometer senses this movement. Changes in the magnetic field components along the x, y, and z axes can be related to angular variation with respect to Earth's magnetic north.

The output signal from the magnetometer is typically an electrical voltage or digital signal that represents the strength and direction of the magnetic field along each axis.

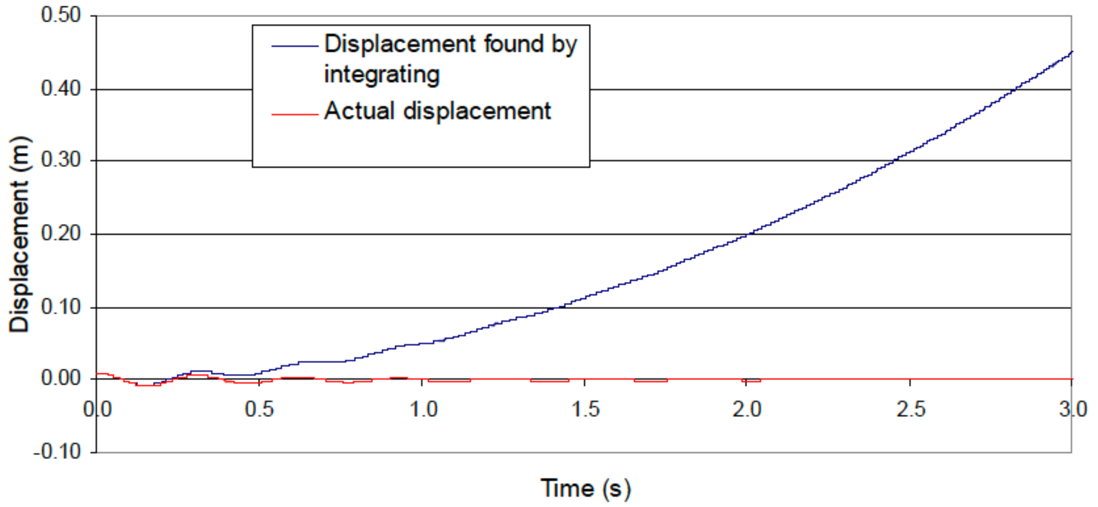
By combining the measurements from all three axes, the magnetometer provides information about the Earth's magnetic field or other magnetic sources in the vicinity.

Inertial sensors

Sensor fusion algorithms estimate orientation by integrating data from the three inertial sensors. The objective of these algorithms is to determine the roll, pitch, and yaw rotation angles of the device within its reference frame. Combining data from different sensors helps mitigate three primary challenges:

- Gyroscope drift
- Magnetic interference
- External acceleration

The double integration of the acceleration provides the change in position, while the integration of the angular velocity provides the rotation angle.
Issue: Accelerometer and gyroscope sensor signals contain errors that make position and orientation estimations challenging due to the integration of drifts.



IMU Sensor Fusion algorithms rely on an orientation estimation filter, such as the Kalman filter.
The Kalman filter is used to combine data from many indirect and noisy measurements. It weights the sources of information based on the knowledge of the signal characteristics and the models that allow the best utilization of the data from each sensor.

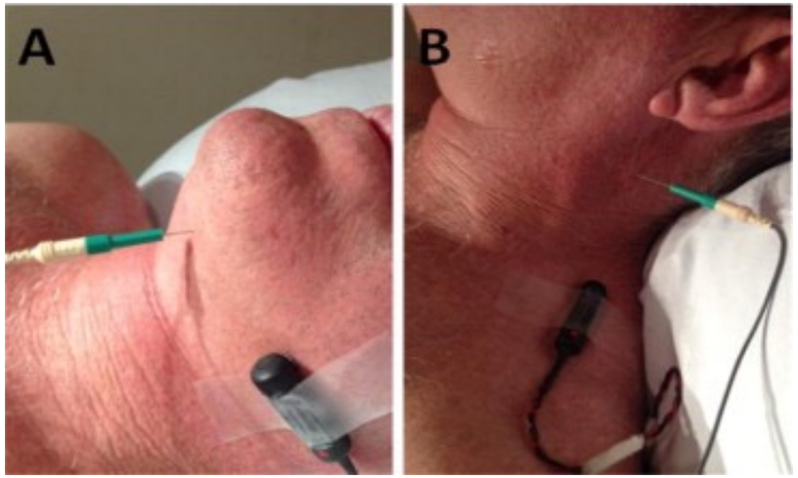
https://www.221e.com/wp-content/uploads/2022/11/221e-Whitepaper-3-IMU-Sensor-Fusion-Troublemakers.pdf

Electromyography

Electromyography (EMG) is a technique used to record and analyze myoelectric signals, which represent the electrical activity of muscles during contraction.

Needle EMG

- Involves inserting a needle electrode into a muscle to record and amplify electrical signals generated by muscle fibers at rest or during contraction.
- The signals are analyzed to assess the functionality of muscle fibers and motor units.
- Commonly applications:
  - Diagnosing and evaluating neuromuscular diseases.
  - Nerve conduction studies: Provides insights into motor unit integrity by recording the summed electrical field generated by muscle fibers on the surface after nerve stimulation.



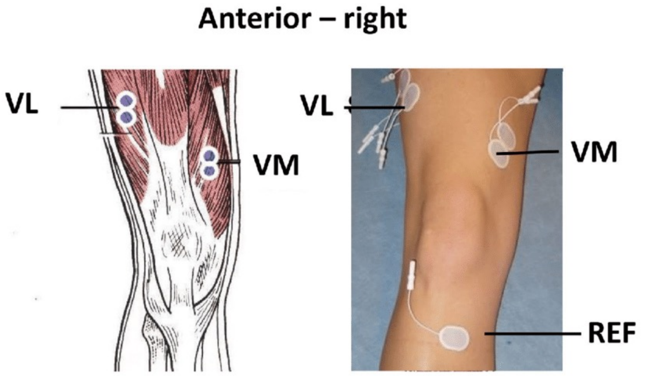
Menkes, D. L., & Pierce, R. (2019). Needle EMG muscle identification: A systematic approach to needle EMG examination. *Clinical neurophysiology practice*, *4*, 199-211.

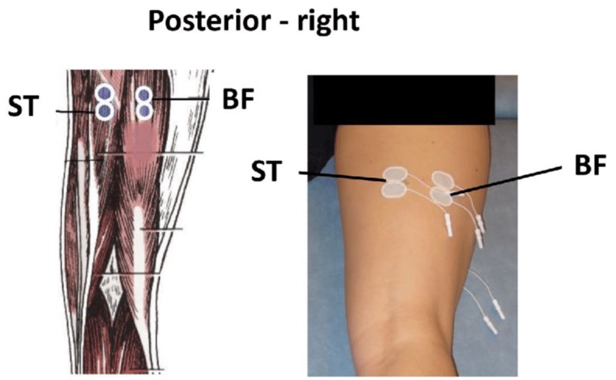
Electromyography

Electromyography (EMG) is a technique used to record and analyze myoelectric signals, which represent the electrical activity of muscles during contraction.

Surface EMG (sEMG)

- Captures electrical activity associated with multiple motor units and is influenced by tissue properties between the electrodes and muscle fibers.
- Applications with high-density surface EMG:
- Grids with more than 32 electrodes placed closely (5-10 mm apart).
    - Identification of single motor unit firing patterns using blind source separation algorithms (e.g., ICA).
    - Validated through simulations and intramuscular EMG recordings.





https://otbioelettronica.it/en/what-is-hd-emg/

Zebis, M. K., Aagaard, P., Andersen, L. L., Hölmich, P., Clausen, M. B., Brandt, M., ... & Bencke, J. (2022). First-time anterior cruciate ligament injury in adolescent female elite athletes: a prospective cohort study to identify modifiable risk factors. *Knee surgery, sports traumatology, arthroscopy*, *30*(4), 1341-1351

Electromyography

Signal Acquisition
- Differential recording to measure the voltage difference between electrodes.
- Use of a reference electrode placed on an electrically neutral area (e.g., bony prominence).

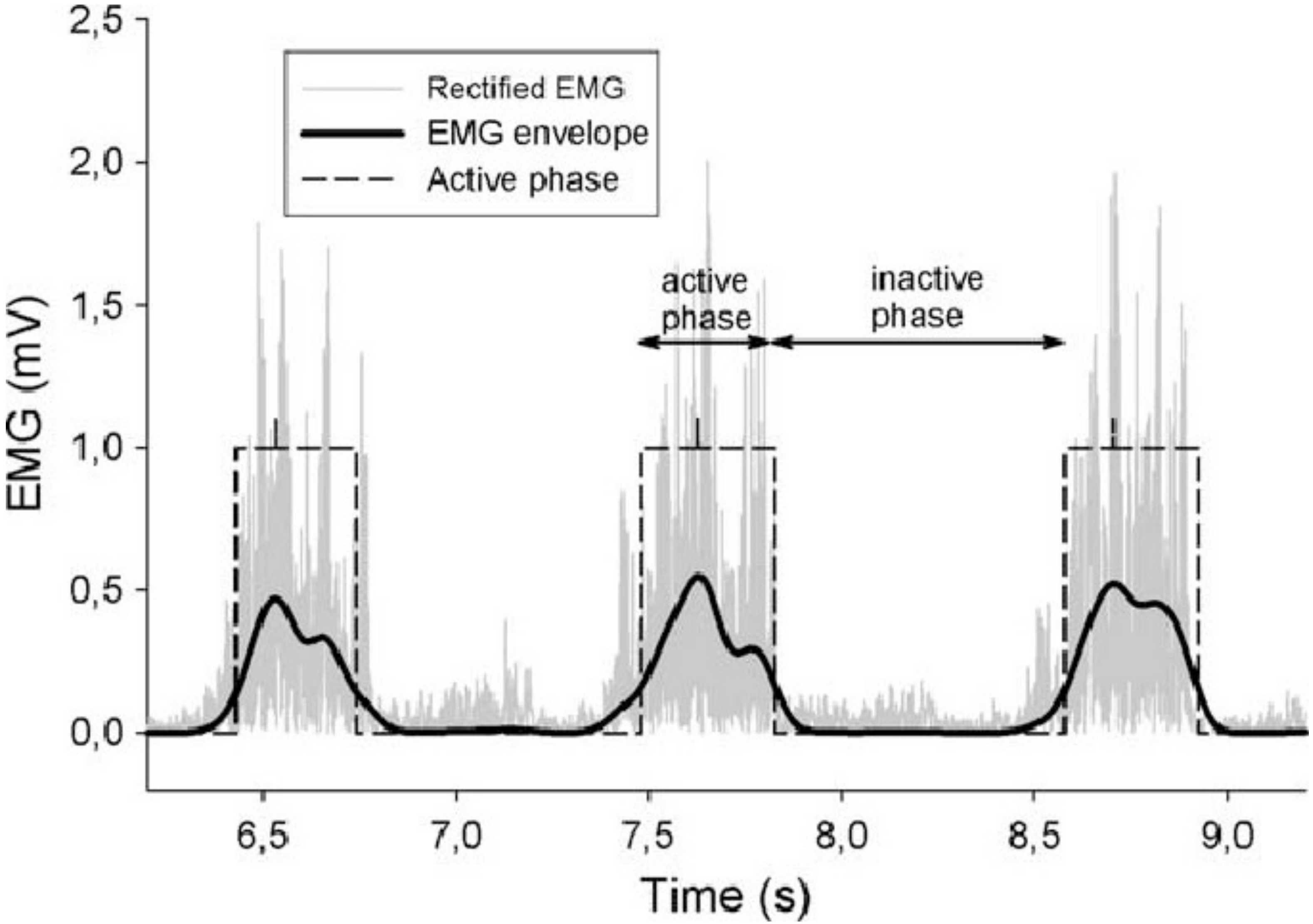
Preprocessing
- High-pass filter (e.g., 5-20 Hz) to remove movement artifacts and baseline drift.
- Low-pass filter (e.g., 500 Hz for sEMG or up to 1500 Hz for needle EMG) to remove high-frequency noise.
- Optional notch filter (50/60 Hz) to eliminate power line interference.

Sampling
- Digital sampling at least 2.5× the low-pass cutoff frequency (e.g., ≥2500 Hz for a 500 Hz cutoff).

Signal Rectification
- Convert raw sEMG signals to their absolute values to preserve amplitude information.

Envelope Extraction
- Low-pass filtering of the rectified signal (cutoff frequency 5-100 Hz) to obtain a smooth envelope.
- Alternative: Use sliding window averaging or digital filter (e.g., Butterworth).

Electromyography



Stirn, I., Jarm, T., Kapus, V. P., & Strojnik, V. (2013). Evaluation of mean power spectral frequency of EMG signal during 100 metre crawl. *European Journal of Sport Science*, *13*(2), 164-173

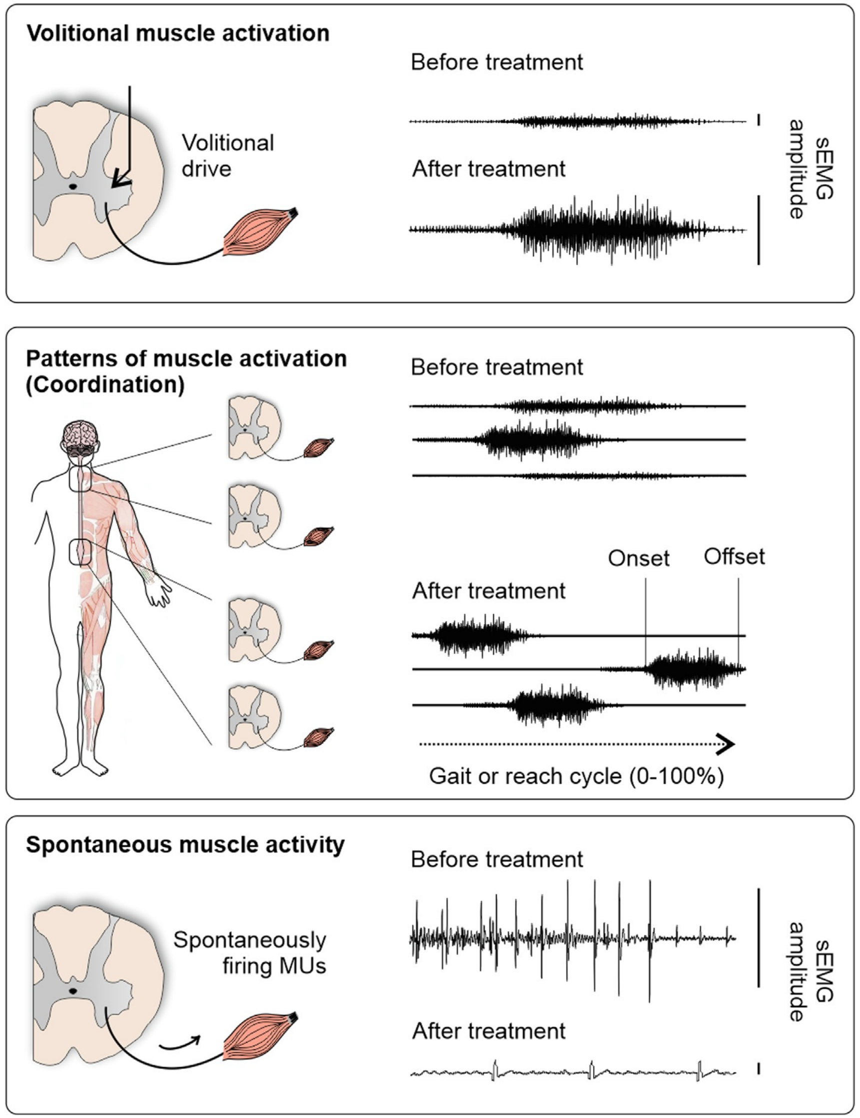
sEMG in neurorehabilitation

- sEMG plays a critical role in assessing the effectiveness of neurorehabilitation strategies, particularly in individuals recovering from spinal cord injury (SCI).

- Neurorehabilitation aims to restore essential motor functions such as standing, walking, reaching, and grasping. Positive outcomes of these interventions can be identified through specific changes in muscle activity patterns, as measured by sEMG.

- These include increased volitional muscle activation (reflected by higher sEMG amplitude), improved coordination among muscle groups (evidenced by rhythmic activity and functional synergies), and a reduction in spontaneous muscle activity, such as spasms or spasticity.

- By quantifying these parameters, sEMG provides valuable insights into the progress and effectiveness of rehabilitation modalities.



Balbinot, G., Wiest, M. J., Li, G., Pakosh, M., Furlan, J. C., Kalsi-Ryan, S., & Zariffa, J. (2022). The use of surface EMG in neurorehabilitation following traumatic spinal cord injury: A scoping review. *Clinical Neurophysiology*, *138*, 61-73.

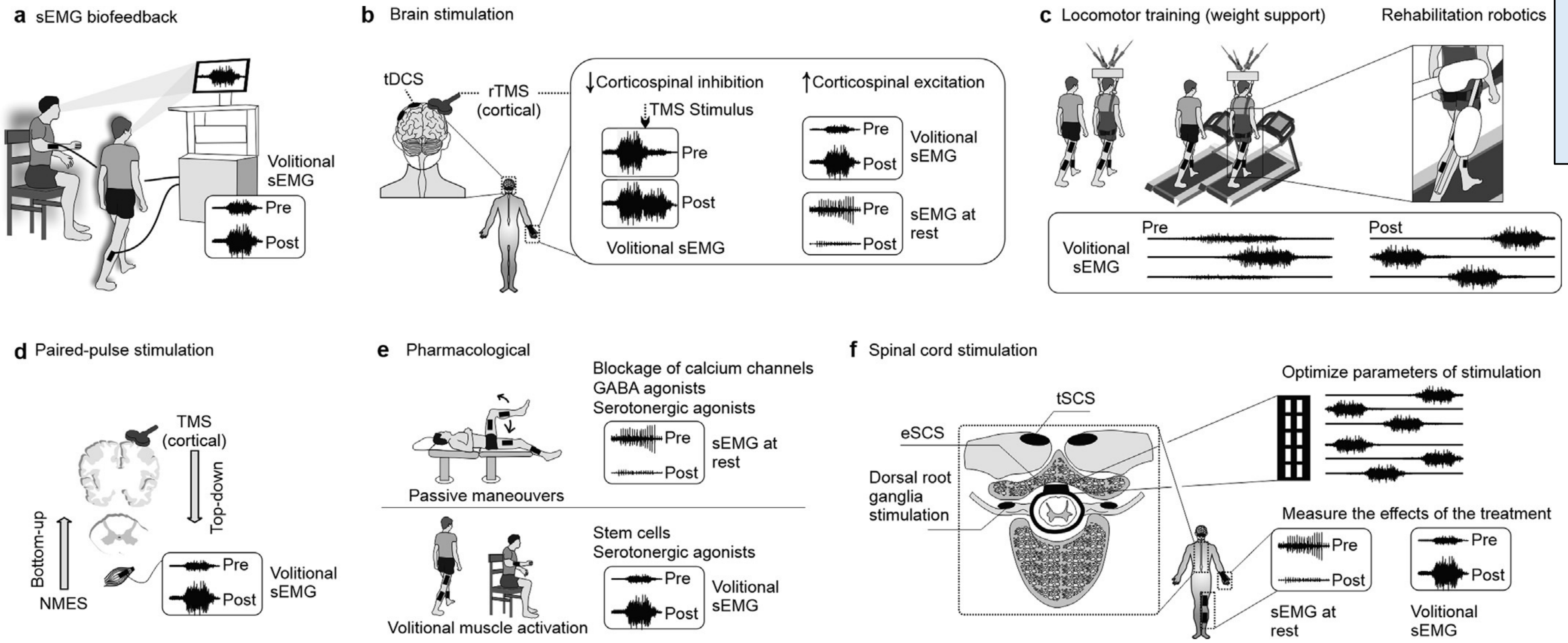
sEMG in neurorehabilitation

sEMG is often used to explore key questions about novel neurorehabilitation strategies after SCI

(a) What are the effects of sEMG biofeedback on volitional control of targeted muscles?

(b) What is the efficacy of brain stimulation in reducing corticospinal inhibition and/or increasing excitation?

(c) What are the neuromuscular changes promoted by different modalities and intensities of locomotor training?



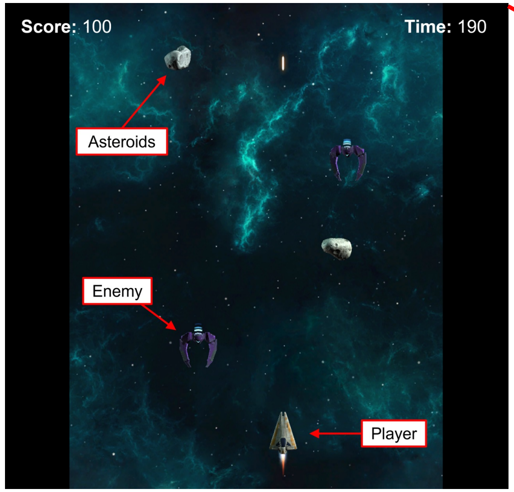
(d) What are the effects of paired-pulse stimulation on corticospinal excitability?

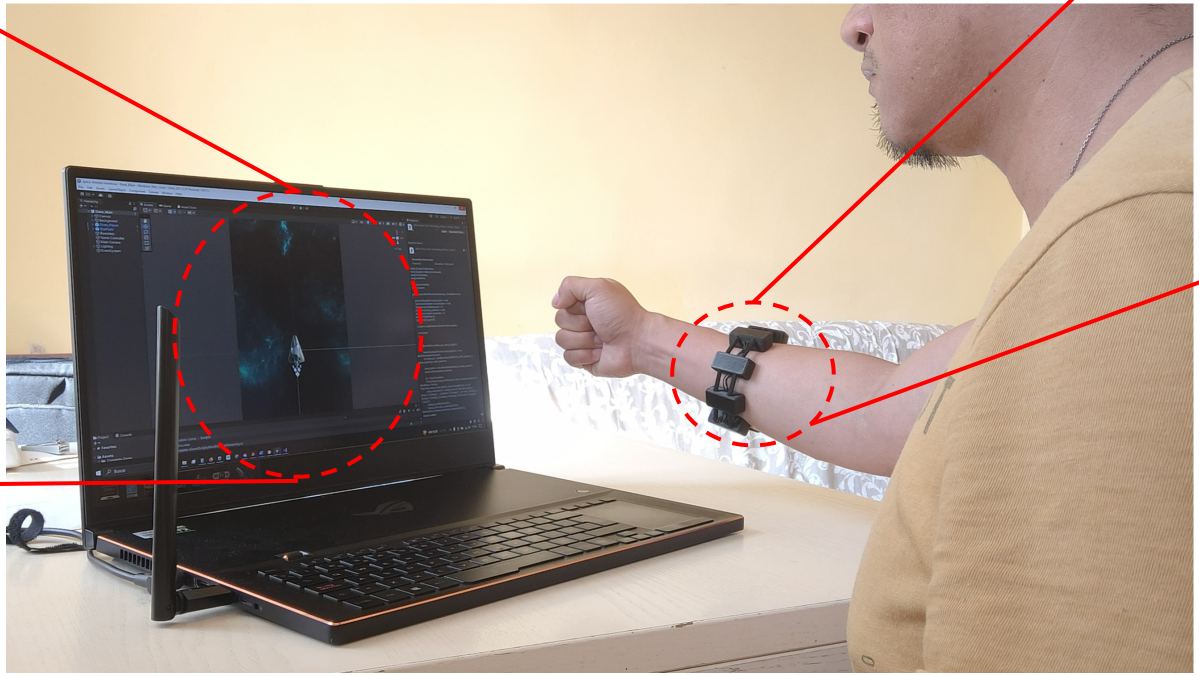
(e) What are the effects of different drugs in reducing persistent spontaneous activity and/or increasing volitional muscle activation?

(f) What are the optimal electrode configurations and stimulation parameters for spinal cord stimulation in individuals living with implanted epidural spinal cord stimulation (eSCS) devices? What are the short- and long-term effects of the stimulation on spontaneous activity and/ or volitional control of muscles?

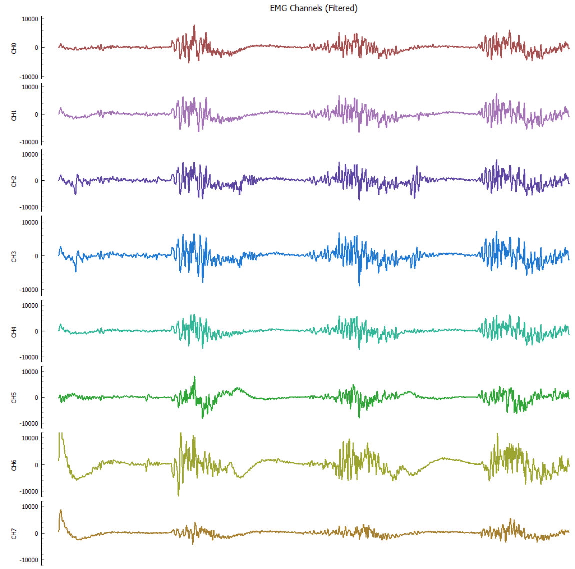
Balbinot, G., Wiest, M. J., Li, G., Pakosh, M., Furlan, J. C., Kalsi-Ryan, S., & Zariffa, J. (2022). The use of surface EMG in neurorehabilitation following traumatic spinal cord injury: A scoping review. *Clinical Neurophysiology*, *138*, 61-73.

Electromyography





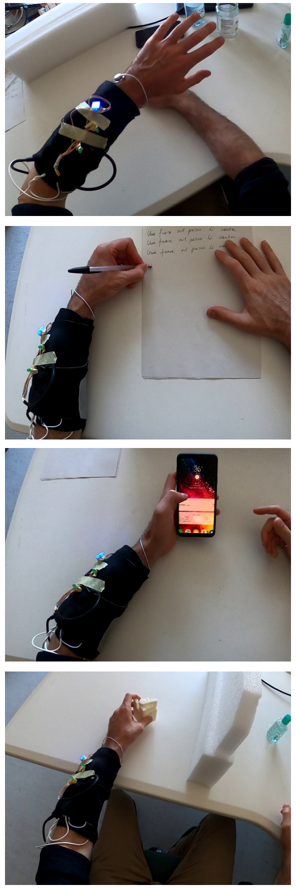
https://mindrove.com/product/armband_8_ch/



Designed for standalone operation. Offers real-time EMG and IMU signal streaming. Electrodes: 8 + 2 stainless steel channels. Sampling Rate: 500 Hz. Includes a 3-axis gyroscope and accelerometer for motion tracking.

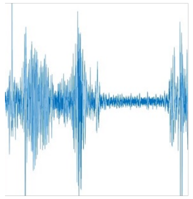
Oña, E. D., Bandini, A., Micera, S., Balaguer, C., & Jardón, A. (2023). EMG-Space Shooter: Integración de Sensor MindRove para Desarrollo de Serious Games Controlados por Activación Electromiográfica del Brazo. In *XLIV Jornadas de Automática* (pp. 83-88). Universidade da Coruña. Servizo de Publicacións.

Electromyography



Titanium carbide ($Ti_3C_2T_x$) MXene, encapsulated in thin layers of flexible parylene-C

OTBioelettronica Sessantaquattro
Fs = 2 kHz



Raw EMG signals

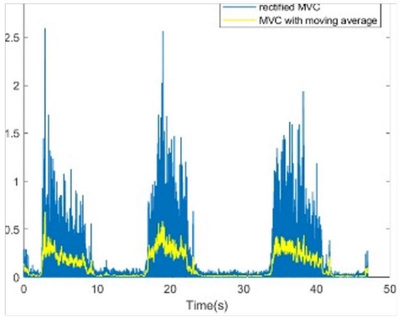
Notch filter (50Hz)

Band-pass filter (10-250Hz)

→

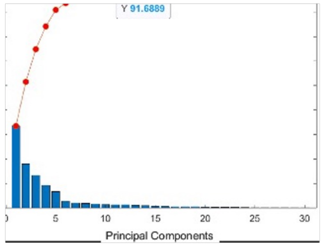
Rectification

Normalization (wrt MVC)



→

PCA (11 PCs – 90% variance)



→

Feature extraction

Time domain

- Integrated EMG (IEMG)
- Average Amplitude Change (AAC)
- Mean Absolute Value (MAV)
- Mean Absolute Deviation (MAD)
- Waveform Length (WL)
- Log Detector (LD)
- Root Mean Square (RMS)
- Variance (VAR)
- Simple Square Integral (SSI)
- The coefficient of variation (COV)
- Kurtosis (KURT)
- Skewness (SKEW)

Frequency domain

- Mean frequency (MNF)
- Median Frequency (MDF)
- Total power (TP)
- Mean Power (MP)
- Peak frequency (PKF)
- Spectral moments (SM1, SM2, SM3)
- Variance of central frequency (VCF)

Features extracted from windows of 1s with 50% overlap

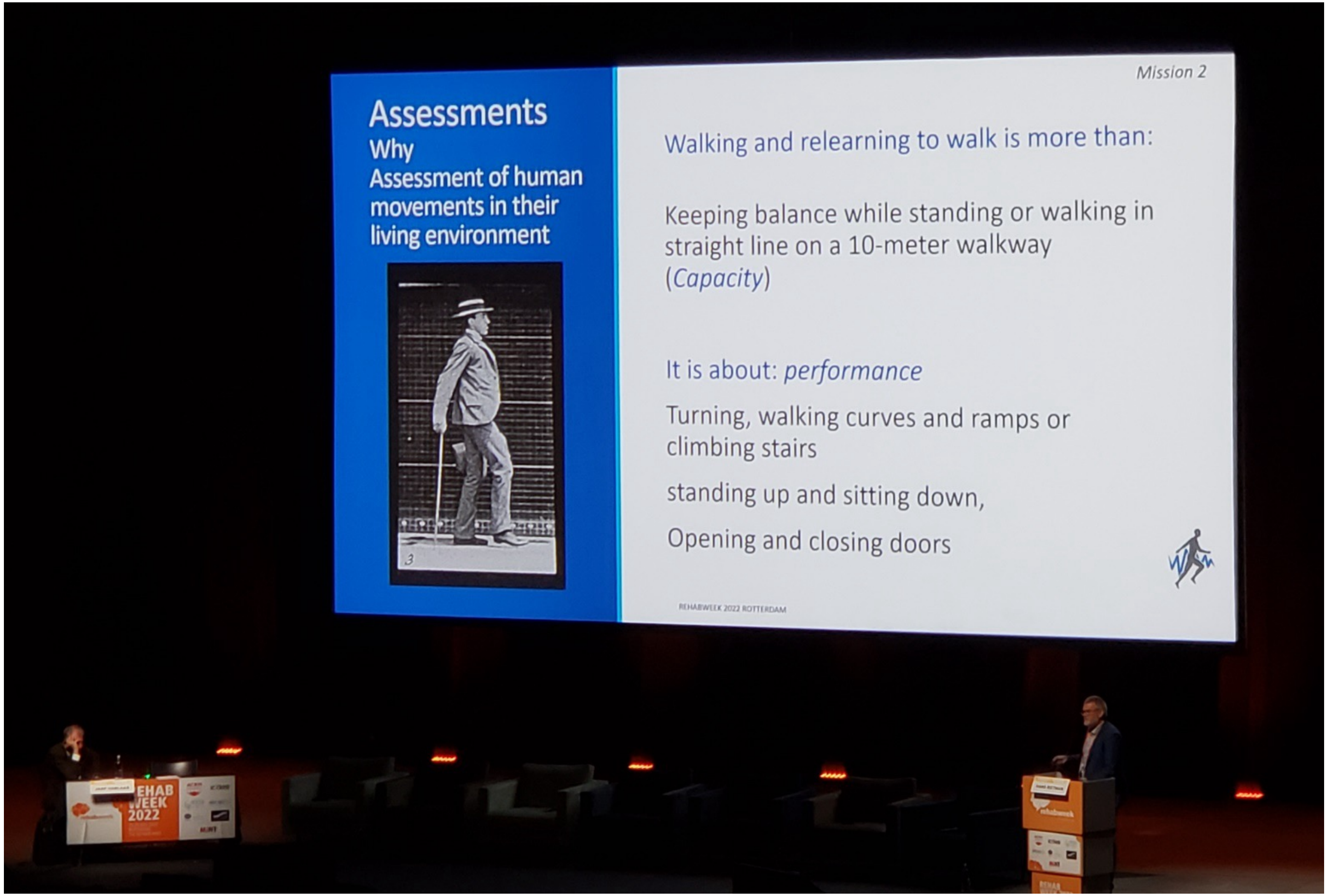
Bandini, A., Zecchin, G., Iberite, F., Proietti, T., Micera, S., & Ambrosini, E. (2023, October). Feasibility of a Portable, Wearable, High-Density Surface EMG Device for Detecting Functional Hand-Object Interactions. In *2023 IEEE International Conference on Metrology for eXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRAINE)* (pp. 846-851). IEEE

3. Technologies for functional assessment

3.1 Clinical assessment
- Overview
- Assessment in neurology and neurorehabilitation

3.2 Technologies for measuring human motion
- Optical motion capture systems
- Wearable sensors
- Electromyography

3.3 Emerging techniques for monitoring and assessment
- Telemonitoring applications
- Prediction of disease course and rehabilitation outcomes



Prof. J.S. (Hans) Rietman, Rehabweek 2022

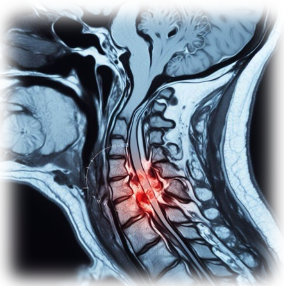
… using the hands and relearning how to use hands is more than:

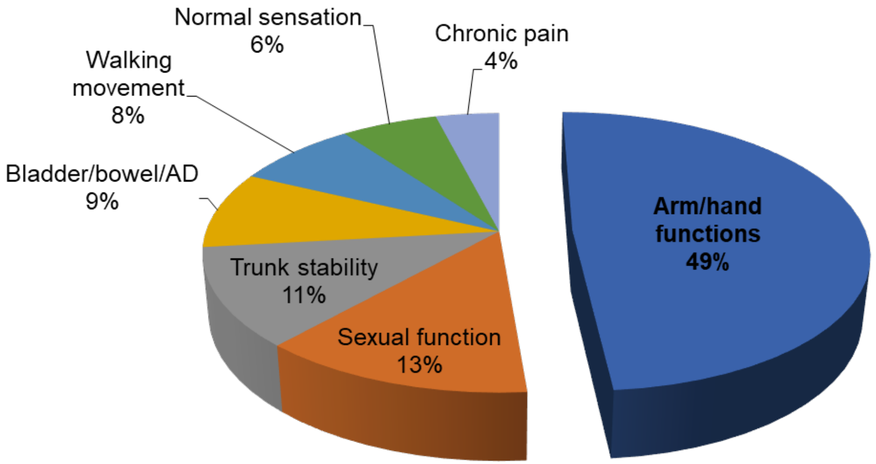
performing reaching and grasping, moving blocks (***capacity***)

it is about:

drinking, eating, preparing meals, dressing (***performance***)

Telemonitoring applications

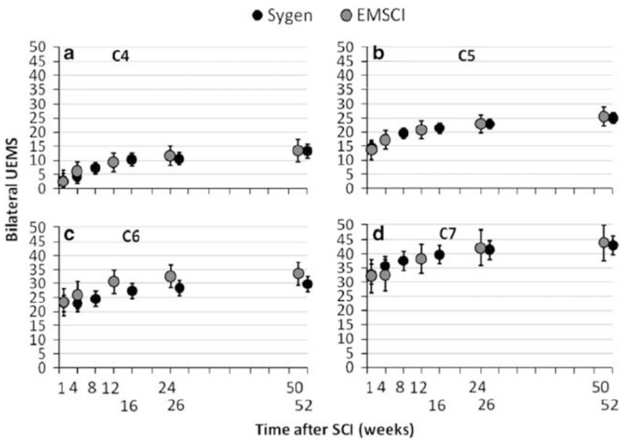




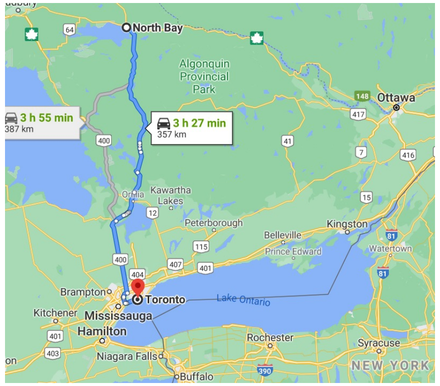
Upper limb function is the top recovery priority for people with cervical spinal cord injury (cSCI)

Barriers to optimal recovery of upper limb functions

1) Patients are discharged too early



2) Distances



3) Lack of outcome measures sensitive to functional hand use

Characteristics of domains of outcome in modified International Classification of Functioning, Disability, and Health model.

Domain	Level of Operation	Example	Can Use Adaptive Aids?	Can Use Another Person's Assistance?
Impairment	Organ system	Strength of finger flexors	No	No
Functional Limitation	Individual as whole	Grasp cylindrical object	No	No
Activity				
Capacity	Individual in standardized environment	Drink from 6 oz glass without handle	Preferably not	Preferably not
Performance	Individual in usual environment	Get drink (any method, i.e., cup, glass, straw)	Yes	Yes
Participation	Individual fulfilling life roles in usual environment	Dine out with friends	Yes	Yes

CAN THEY?

DO THEY?

Anderson, K. D. (2004). Targeting recovery: priorities of the spinal cord-injured population. *Journal of neurotrauma*, *21*(10), 1371-1383
Marino, R. J. (2007). Domains of outcomes in spinal cord injury for clinical trials to improve neurological function. *Journal of Rehabilitation Research & Development*, *44*(1)

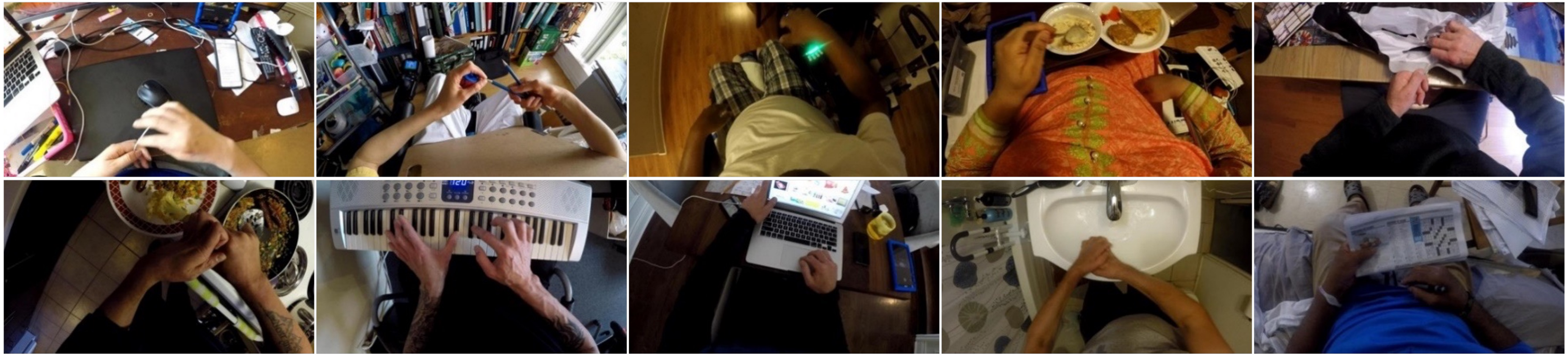
Telemonitoring applications

The ideal technology for monitoring the functional use of the hands at home should:

- Track the rehabilitation progress remotely
- Measure increased hand use in daily life
- Provide measures for planning and validating interventions

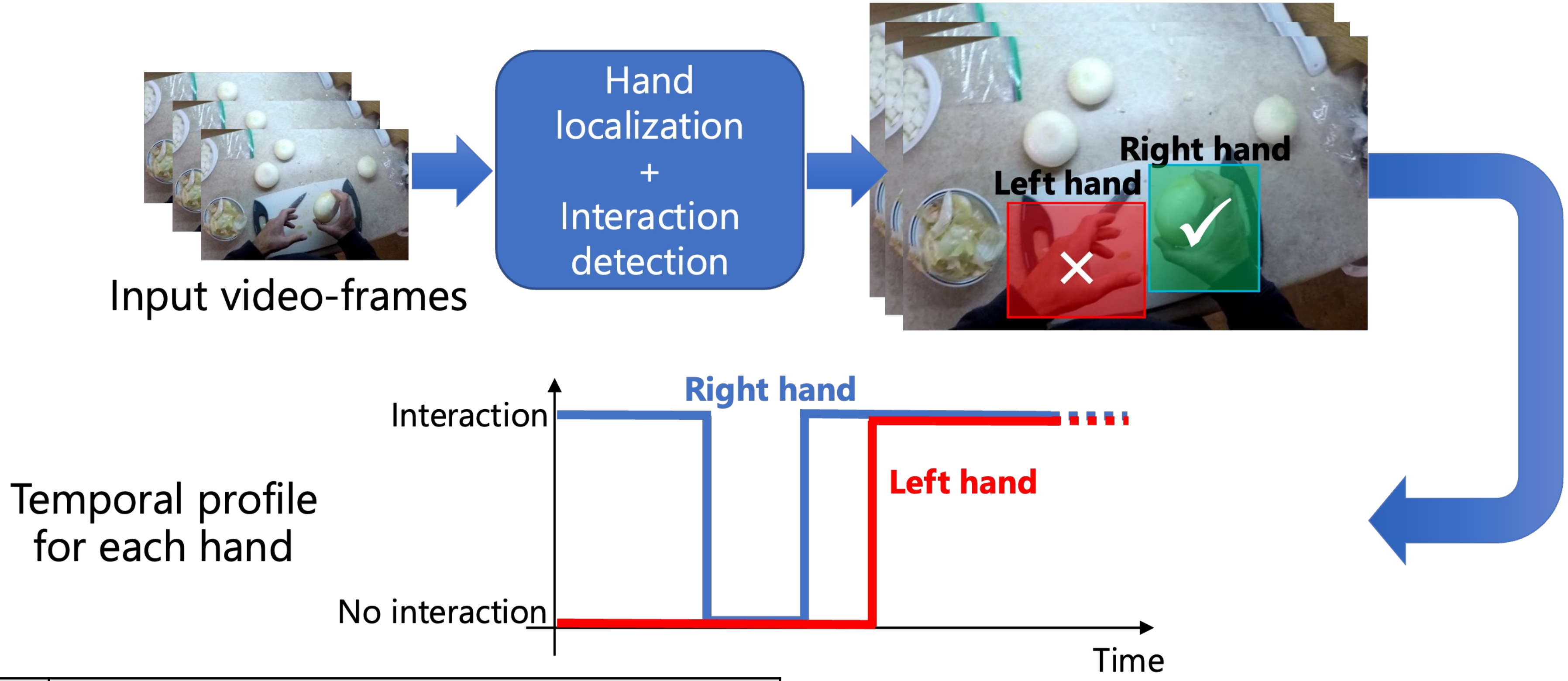
Egocentric vision (i.e., cameras worn on the head)

- Camera movement driven by the user's attention
- Recording exactly what the users have in front of them

Bandini, A., & Zariffa, J. (2023). Analysis of the Hands in Egocentric Vision: A Survey. *IEEE Transactions on Pattern Analysis & Machine Intelligence*, *45*(06), 6846-6866
Bandini, A., Dousty, M., Hitzig, S. L., Craven, B. C., Kalsi-Ryan, S., & Zariffa, J. (2022). Measuring hand use in the home after cervical spinal cord injury using egocentric video. *Journal of neurotrauma*, *39*(23-24), 1697-1707.

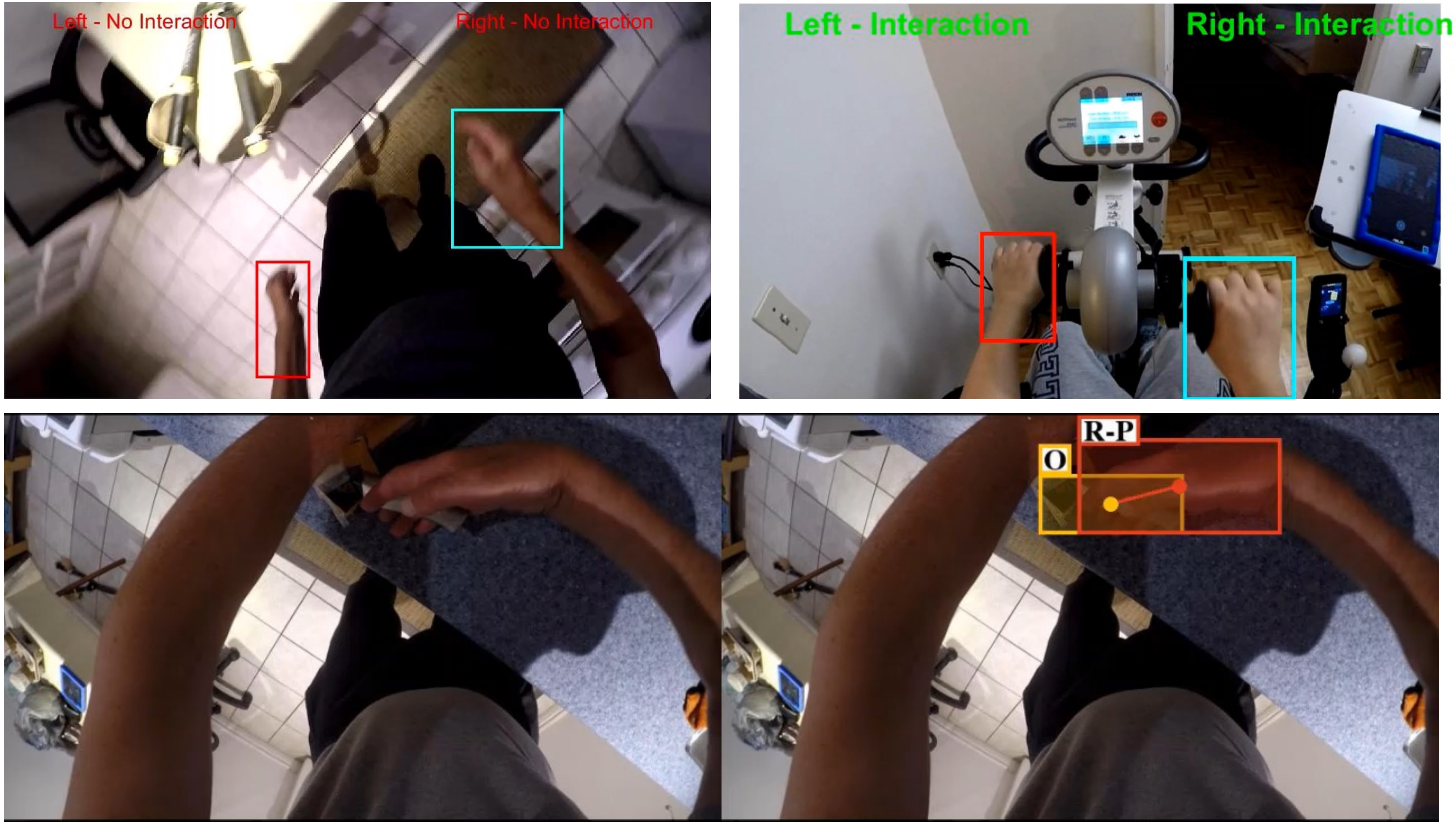
Telemonitoring applications



Perc (%)	The amount of total interaction time as a percentage of the recording time
Dur (s)	The average duration of individual interactions
Num	The number of interactions per hour

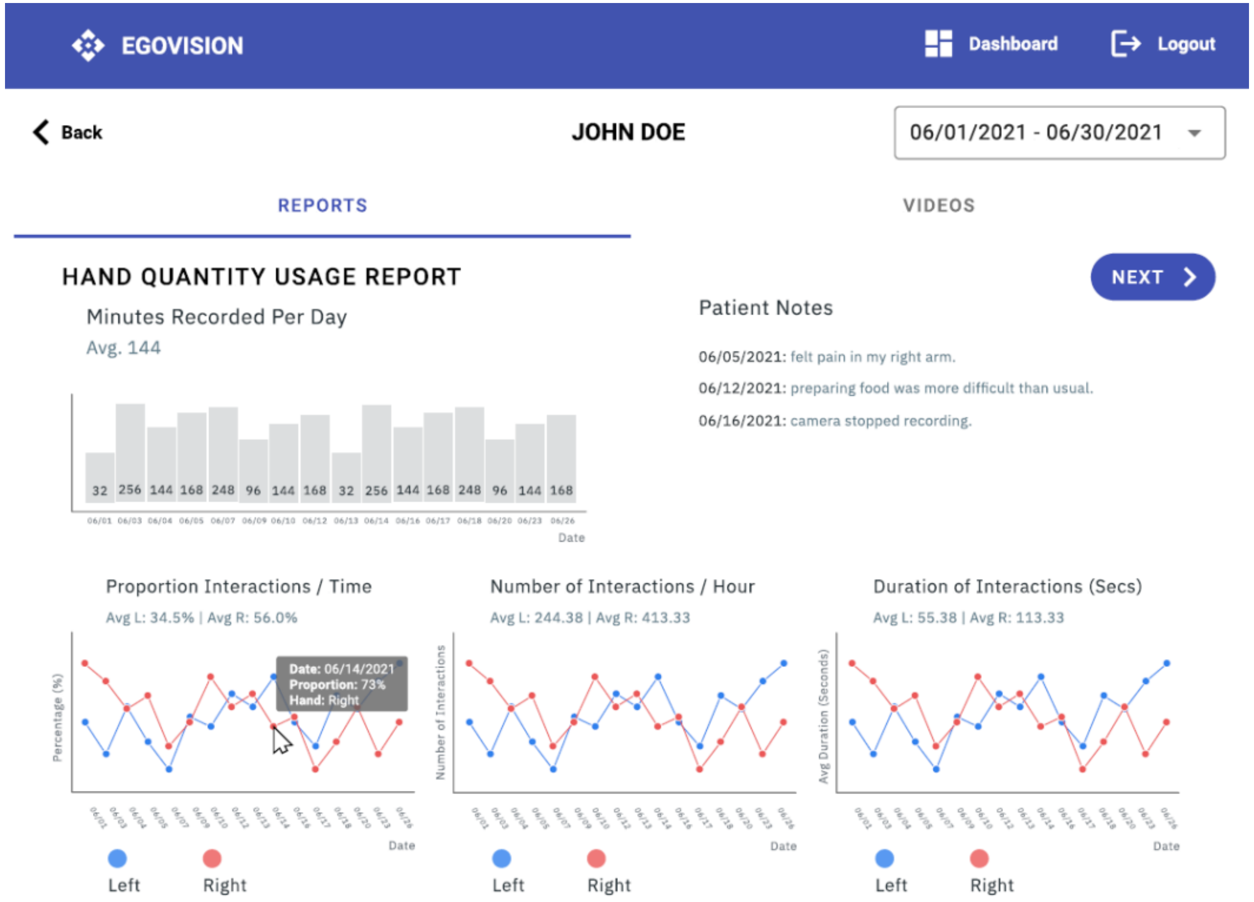
Bandini, A., Dousty, M., Hitzig, S. L., Craven, B. C., Kalsi-Ryan, S., & Zariffa, J. (2022). Measuring hand use in the home after cervical spinal cord injury using egocentric video. *Journal of neurotrauma*, *39*(23-24), 1697-1707.

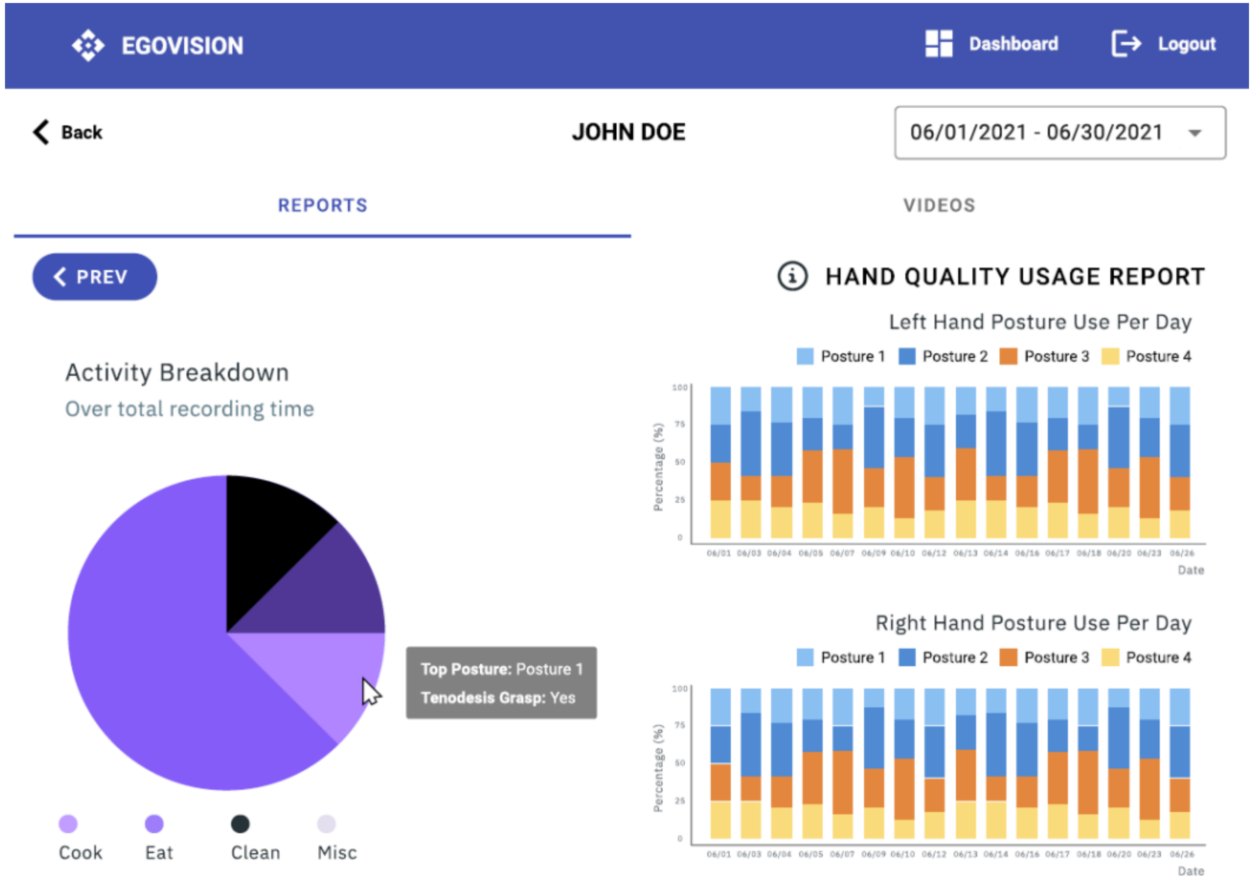
Telemonitoring applications



Bandini, A., Dousty, M., Hitzig, S. L., Craven, B. C., Kalsi-Ryan, S., & Zariffa, J. (2022). Measuring hand use in the home after cervical spinal cord injury using egocentric video. *Journal of neurotrauma*, *39*(23-24), 1697-1707.

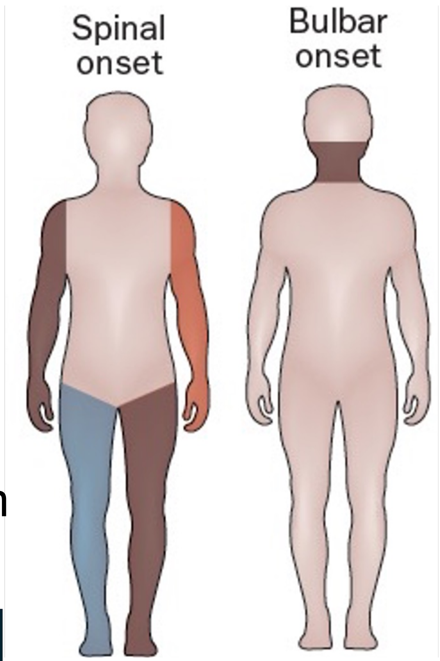
Telemonitoring applications





Kadambi, A., Bandini, A., Ramkalawan, R. D., Hitzig, S. L., & Zariffa, J. (2023). Designing an Egocentric Video-Based Dashboard to Report Hand Performance Measures for Outpatient Rehabilitation of Cervical Spinal Cord Injury. *Topics in Spinal Cord Injury Rehabilitation*, *29*(Supplement), 75-87.

Prediction of disease course



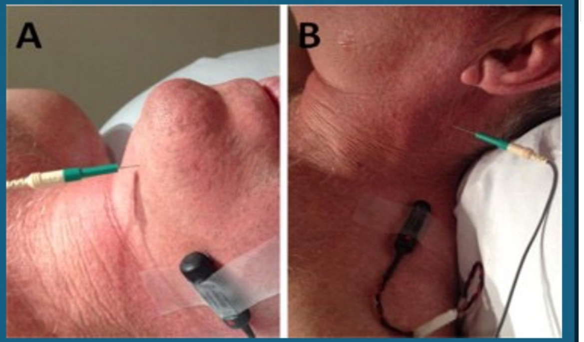
Amyotrophic lateral sclerosis (ALS) is a progressive neurodegenerative disease that affects motor neurons, leading to paralysis and eventual death.

Bulbar symptoms (speech and swallowing impairments)

- Difficulties in swallowing, chewing, speaking, and breathing
- At onset: 30% of cases; most patients with spinal onset will eventually develop them
- Shorter survival rate (<2 years) and more debilitating disease course

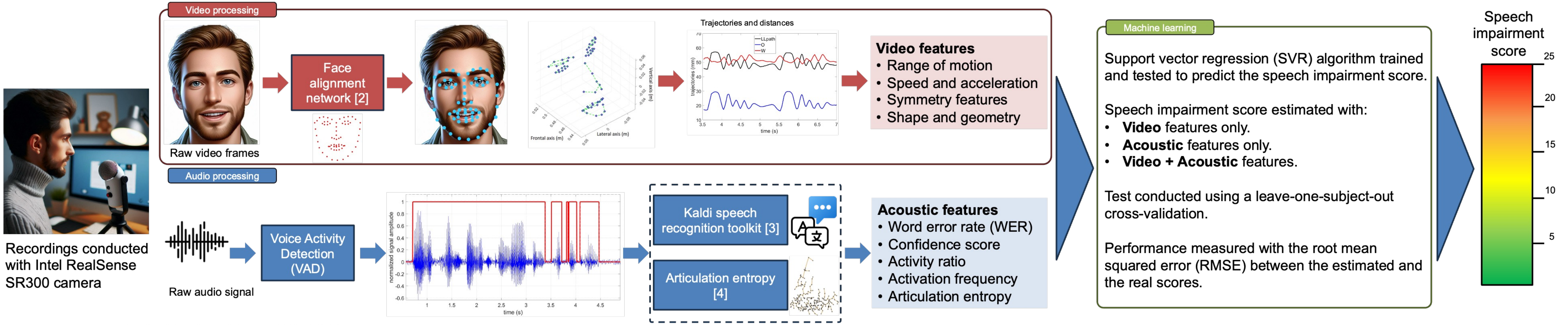
Current assessment tools, (e.g., ALS Functional Rating Scale–Revised – ALSFRS-R) and clinical judgments of speech quality, are inadequate

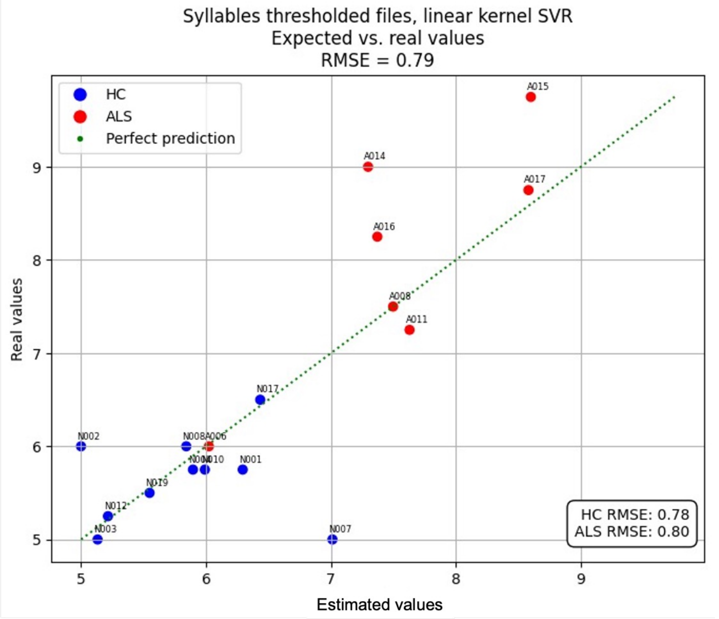
EMG is difficult to perform on bulbar muscles (e.g., tongue and laryngeal mucles)



The objective is to perform an **early detection and prediction of bulbar decline**, which is crucial for improving patient care, including communication and nutrition support, and life-prolonging interventions.

Prediction of disease course





Audio features - Best results obtained with a linear SVR model on syllables (RMSE = 0.79).
Video features - Best results obtained with a sigmoidal SVR model on sentences (RMSE = 1.57).
Audio+Video features - Best results obtained with a radial basis function SVR (RMSE = 1.36).

- Acoustic data provides the most accurate assessment of bulbar impairment in ALS.
- Visual speech data has supplementary value but is less precise on its own.
- Combining multimodal features shows promise but requires further refinement to outperform audio models.

Clinical Relevance

- The findings underscore the utility of acoustic and visual speech analysis as non-invasive tools for monitoring bulbar decline in ALS.
- These preliminary results provide a foundation for developing automated diagnostic systems to assist clinicians in evaluating disease progression.
- These results provide a foundation for future longitudinal studies aimed at automating the assessment of bulbar impairment and predicting its progression over time.

Prediction of disease course